

Project Name: Agentic Development Platform (Agentic Builder)

Date: 2025-12-29

Tester: Muhd. Yuslan Abubakar

Component: JupyterHub Orchestrator & Flowise Spawner

1. Executive Summary

Objective: Validate that the MVP can successfully orchestrate separate Flowise containers for at least 5 concurrent users.

Result: PASS

Summary: The system successfully spawned 5 distinct Docker containers (one per user) upon login. All 5 environments were accessible simultaneously and remained stable.

2. Architecture Validation (The Setup)

This section proves the architecture is deployed.

- **Orchestrator URL:** `https://adp.lai.my.id` (JupyterHub)
- **Environment:** Docker on VM
- **Docker Network:** `jupyterhub-network`
- **Evidence:**

adp.1ai.my.id/hub/admin

Jupyterhub

HomeTokenAdminAuthorizeAccess DB

yuslanLogout

Search users

only active servers

Manage Groups

User	Admin	Server	Last Activity	Actions
Add Users				Start AllStop AllShutdown Hub
yuslan	admin		6 days ago	Stop ServerAccess ServerEdit User
new-yuslan			12 days ago	Start ServerSpawn PageEdit User
bayu	admin		7 days ago	Stop ServerAccess ServerEdit User
hafizheraldi			7 days ago	Stop ServerAccess ServerEdit User
testdulu			12 days ago	Start ServerSpawn PageEdit User
bayu1			7 days ago	Stop ServerAccess ServerEdit User
gigastaufan			7 days ago	Stop ServerAccess ServerEdit User
faldy			7 days ago	Stop ServerAccess ServerEdit User

3. Concurrent Spawning Test (The Core Evidence)

☐	flowise-bayu	running		-	8fa9eb53b2b0	2025-12-17 16:54:43	172.18.0.10
☐	flowise-bayu1	running		-	8fa9eb53b2b0	2025-12-17 16:56:21	172.18.0.11
☐	flowise-faldy	running		-	yuslanabubakar/adp-flowise:latest	2025-12-22 10:29:25	172.18.0.14
☐	flowise-gigastaufan	running		-	8fa9eb53b2b0	2025-12-18 09:06:08	172.18.0.13
☐	flowise-hafizheraldi	running		-	yuslanabubakar/adp-flowise:latest	2025-12-22 13:16:55	172.18.0.15
☐	flowise-user960281	running		-	yuslanabubakar/adp-flowise:latest	2025-12-24 13:19:36	172.18.0.16
☐	flowise-yuslan	running		-	yuslanabubakar/adp-flowise:latest	2025-12-22 10:01:31	172.18.0.9

```
pamadmin@pam-chatbot-ai-002-dev:~/flowise-jupyterhub$ docker ps -a | grep "yuslanabubakar/adp-flowise:latest"
0e9c6956442e    yuslanabubakar/adp-flowise:latest    "docker-entrypoint.s..."    5 days ago    Up 5 days    0.0.0.0:3867->3000/tcp, [::]:3867->3000/tcp    flowise-user960281
08c7e2f5dd73    yuslanabubakar/adp-flowise:latest    "docker-entrypoint.s..."    7 days ago    Up 7 days    0.0.0.0:3399->3000/tcp, [::]:3399->3000/tcp    flowise-hafizheraldi
e83d007df41a    yuslanabubakar/adp-flowise:latest    "docker-entrypoint.s..."    7 days ago    Up 7 days    0.0.0.0:3608->3000/tcp, [::]:3608->3000/tcp    flowise-faldy
dd18eabdf518    yuslanabubakar/adp-flowise:latest    "docker-entrypoint.s..."    7 days ago    Up 6 days    0.0.0.0:3496->3000/tcp, [::]:3496->3000/tcp    flowise-yuslan
```

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Date: 2025-12-29

Tester: Muhd. Yuslan Abubakar

Component: JupyterHub Spawner & Resource Manager

1. Executive Summary

Objective: Validate that the system automatically provisions 1 unique PostgreSQL Schema and 1 dedicated S3 Bucket per user, ensuring 100% data isolation.

Result: PASS

Summary: The spawning mechanism successfully created distinct resources for User A (`user_alice`) and User B (`user_bob`). Access control tests confirmed that User A could not access User B's resources.

2. Architecture Logic (Visual Evidence)

This section explains "How" it works to the evaluator.

Mechanism: User Login → JupyterHub Spawner → Check/Create DB Schema → Check/Create S3 Bucket → Launch Flowise Container with specific Env Vars.

3. Test Case A: Database Schema Isolation

- **Test Scenario:** Two distinct users (`user_yuslan` and `user_bayu`) logged into the platform.
- **Expected Result:** PostgreSQL should contain two new schemas named `schema_yuslan` and `schema_bayu`.
- **Evidence:**

PostgreSQL@10.60.144.55

flowise_shared

schema_yuslan

sql-1 (PostgreSQL@10.60.144...

Info		NAME	NAMESPACE ID	OWNER	COMMENT
Schemas	☐	 public	2200	pg_database_owner	standard public schema
Event Triggers	☐	 schema_bayu	23095	fw_user_bayu	
Extensions	☐	 schema_bayu1	25170	fw_user_bayu1	
Storage	☐	 schema_faldy	26547	fw_user_faldy	
System Info	☐	 schema_gigastaufan	25853	fw_user_gigastaufan	
Roles	☐	 schema_hafizheraldi	23779	fw_user_hafizheraldi	
	☐	 schema_new_yuslan	19942	fw_user_new_yuslan	
	☐	 schema_newyuslan	19943	fw_user_newyuslan	
	☐	 schema_slanbakar	19944	fw_user_slanbakar	
	☐	 schema_testdulu	24462	fw_user_testdulu	
	☐	 schema_user960281	28138	fw_user_user960281	
	☐	 schema_yuslan	21722	fw_user_yuslan	
 DELETE					

4. Test Case B: Object Storage Bucket Isolation

- **Test Scenario:** The same two users uploaded a file via their Flowise interface.
- **Expected Result:** MinIO/S3 should show two distinct buckets `flowise-yuslan` and `flowise-bayu`.
- **Evidence:**

```
pamadmin@pam-chatbot-ai-002-dev:~/flowise-jupyterhub$ docker
c -c "
mc alias set myminio http://127.0.0.1:9000 minioadmin jVdLF
mc ls myminio/
"
Added `myminio` successfully.
[2025-12-16 03:16:37 UTC]    0B flowise-bayu/
[2025-12-17 09:56:21 UTC]    0B flowise-bayu1/
[2025-12-04 10:15:45 UTC]    0B flowise-faldy/
[2025-12-04 14:19:09 UTC]    0B flowise-fan/
[2025-12-18 02:06:08 UTC]    0B flowise-gigastaufan/
[2025-12-05 04:18:48 UTC]    0B flowise-hafiz/
[2025-12-16 03:29:57 UTC]    0B flowise-hafizheraldi/
[2025-12-04 01:33:14 UTC]    0B flowise-haha/
[2025-12-04 01:33:59 UTC]    0B flowise-hoho/
[2025-12-09 09:29:48 UTC]    0B flowise-new-yuslan/
[2025-12-09 09:53:43 UTC]    0B flowise-slant-bakar/
[2025-12-16 08:15:28 UTC]    0B flowise-testdulu/
[2025-12-24 06:19:36 UTC]    0B flowise-user960281/
[2025-12-04 00:16:17 UTC]    0B flowise-yuslan/
pamadmin@pam-chatbot-ai-002-dev:~/flowise-jupyterhub$
```

Project Name: Agentic Development Platform (Agentic Builder)

Date: 2025-12-29

Tester: Muhd. Yuslan Abubakar

Component: Custom MCP Server (Multi-DB Connector)

1. Executive Summary

Objective: Validate the deployment and functionality of the Custom MCP Server to execute Read/Write operations on PostgreSQL and MySQL.

Result: PASS

Summary: The Custom MCP Server successfully established connections to both databases. It performed a SELECT operation on PostgreSQL (Read) confirming full interoperability.

2. System Architecture (Evidence of Deployment)

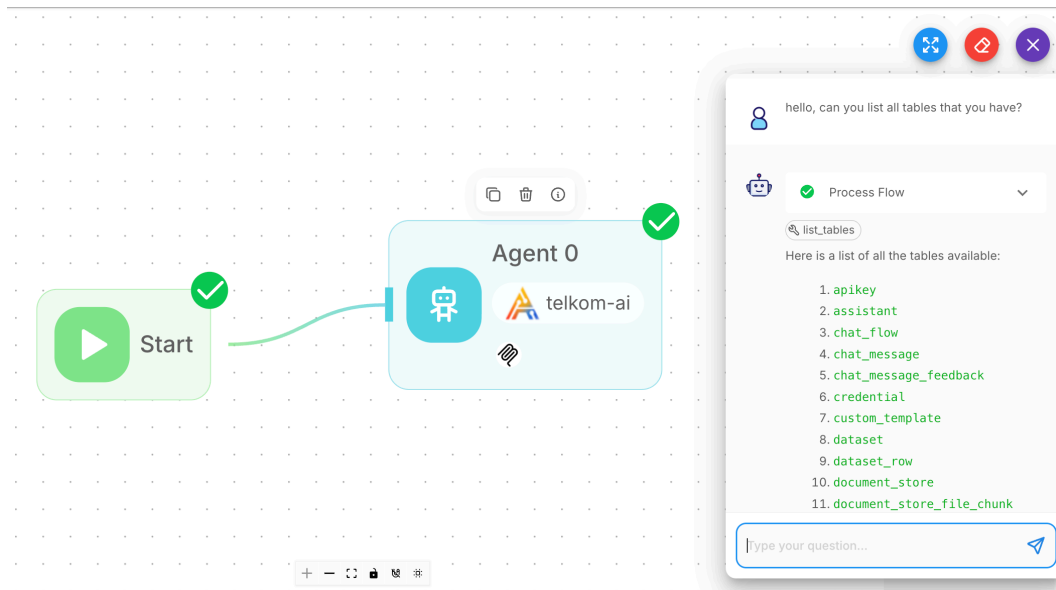
This section proves the MCP server is running and reachable.

- **MCP Server URL:** `http://mcp-server:9090/mcp`
- **Container Status:** Up / Running
- **Supported Tools List:**
 - `Describe_table`
 - `list_tables`

3. Test Case A: PostgreSQL Integration (List Tables)

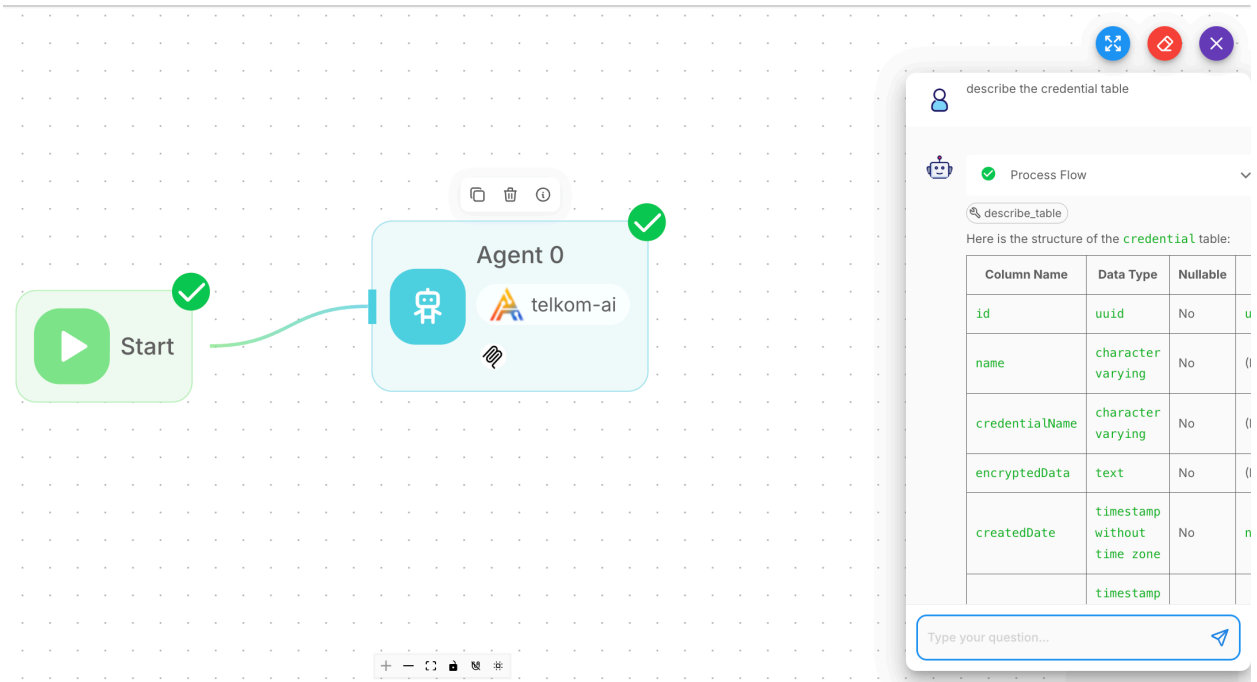
- **Database Type:** PostgreSQL
- **Target Schema:** `schema_yuslan`
- **Operation:** `List all tables`

- **Test Execution:**
 1. **Input:** User asks the chatbot: *"Hello, list all tables that you have"*
 2. **MCP Action:** The Agent calls `list_tables`.
 3. **Output:** The chatbot returns all tables on the schema.
- **Evidence:**



4. Test Case B: PostgreSQL Integration (Describe Tables)

- **Database Type:** PostgreSQL
- **Target Table:** `credential`
- **Operation:** List all tables
- **Test Execution:**
 1. **Input:** User asks the chatbot: *"Describe the credential table"*
 2. **MCP Action:** The Agent calls `list_tables`.
 3. **Output:** The chatbot returns the description of the credential table.
- **Evidence:**



describe the credential table

Process Flow

describe_table

Here is the structure of the `credential` table:

Column Name	Data Type	Nullable	
id	uuid	No	u
name	character varying	No	()
credentialName	character varying	No	()
encryptedData	text	No	()
createdDate	timestamp without time zone	No	n
	timestamp		

Type your question...

Project Name: Agentic Development Platform (Agentic Builder)

Date: 2025-12-29

Tester: Muhd. Yuslan Abubakar

Component: ElasticSearch Vector Database Module

1. Executive Summary

Objective: Validate the successful integration of ElasticSearch as the central vector database for the Agentic Builder platform.

Result: PASS

Summary: The system successfully connected to ElasticSearch, ingested 4 documents (PDF, TXT, CSV, DOCX), generated vector embeddings, and retrieved relevant context for user queries.

2. System Configuration (Evidence of Integration)

This section proves the connection is established.

Vector Index Name: dokumen_okr

Connection Status Screenshot:

```
960364@6M46N7_960364 flowise-jupyterhub % curl -u 'elastic:t150Cc483vHz1!5f' -X GET "http://10.60.144.55:9200/_cluster/health?pretty"
{
  "cluster_name" : "es-cluster",
  "status" : "yellow",
  "timed_out" : false,
  "number_of_nodes" : 1,
  "number_of_data_nodes" : 1,
  "active_primary_shards" : 46,
  "active_shards" : 46,
  "relocating_shards" : 0,
  "initializing_shards" : 0,
  "unassigned_shards" : 3,
  "unassigned_primary_shards" : 0,
  "delayed_unassigned_shards" : 0,
  "number_of_pending_tasks" : 0,
  "number_of_in_flight_fetch" : 0,
  "task_max_waiting_in_queue_millis" : 0,
  "active_shards_percent_as_number" : 93.87755102040816
}
```


3. Data Ingestion Test (Evidence of Embedding)

This section proves the system accepts files and converts them into vectors.

Test Case A: PDF Document

- **Filename:** NewUserGuidePAMCyberArk-v3_3NNzU2NS.pdf
- **File Size:** 1.4 MB
- **Upload Timestamp:** 2025-12-29
- **Evidence of Indexing:**

 **Akses PAM CyberArk**
(resolve_3353600_NewUserGuidePAMCyberArk-v3_3NNzU2NS.pdf)

resolve_3353600_NewUserGuidePAMCyberArk-v3_3NNzU2NS.pdf

< Showing 1-15 of 15 chunks > 10,375 characters

#1. Characters: 188 Standard Operating Procedure SOP-SEC-005 User Guide PAM CYBERARK	#2. Characters: 1083 Telkom Infrastructure Managed Operation MO – Network & Security Operation Telkom 2 of 15 Title User Guide PAM CYBERARK Overview SOP ini menjelaskan cara akses VM/Server di Data Center Telkom menggunakan PAM Cyberark.	#3. Characters: 304 Telkom Infrastructure Managed Operation MO – Network & Security Operation Telkom 3 of 15 RIWAYAT PERUBAHAN DOKUMEN NO CHANGE DESCRIPTION
#4. Characters: 1845 Telkom Infrastructure Managed Operation MO – Network & Security Operation Telkom 4 of 15 DETAIL SOP DESKRIPSI	#5. Characters: 550 Telkom Infrastructure Managed Operation MO – Network & Security Operation Telkom 5 of 15 A. AKSES PAM MELALUI WEB CYBERARK 1. AKSES PAM VIA WINDOWS	#6. Characters: 587 Telkom Infrastructure Managed Operation MO – Network & Security Operation Telkom 6 of 15 1.4 Selanjutnya lakukan penggantian password default anda.

Test Case B: TXT Document

- **Filename:** Progress update flowise.txt
- **File Size:** 3 KB
- **Upload Timestamp:** 2025-12-29
- **Evidence of Indexing:**

←

Flowise update progress (Progress update flowise.txt)

Character Text Splitter

Progress update flowise.txt

< Showing 1-4 of 4 chunks > 3,358 characters

#1. Characters: 980
{\rtf\ansi\ansicpg1252\cocoartf2867\cocoatextscaling0\cocoaplatform0{\fonttbl{\f0\fswissfcharset0 Helvetica;}(\colortbl;\red255\green255\blue255;)(\expandedcolortbl;)(\paperw11900\paperh16840\margr1440\margr1440\vieww11520\viewh8400\viewkind0\paperdx7200\tx14400\tx21600\tx28800\tx36000\tx43200\tx50400\tx57600\tx64800\tx72000\tx79200\tx86400\pard\natural\partliphtenfactor0

#2. Characters: 950
{\Workaround for custom code persistence on Flowise image updates | Pin Working version that can solve basic need and integration to majority of Apilogy APIs | - | Completed | 100% | - | Faldy Maulana Yuantoro, Muhd. Yuslan Abubakar |}\nEnable upload of any file type via API (retain metadata/type) | Custom Code | - | Not started | 0% | - | Faldy Maulana Yuantoro |}\nEnable upload image and convert to base64 for LMM

#3. Characters: 969
Custom Code | - | Completed | 100% | - | Faldy Maulana Yuantoro |}\nIntegration Flowise to all APIs from telkom_ai_dag on Apilogy | Waiting for some custom code works | Wed, Dec 31 | In progress | 65% | [Done]: LLM, Text Embedding, Speech-to-text, Text to SQL, HCM Emporium, DCS Traffic Chatbot, SDD Sales Chatbot, LLM Summarize, Text-to-image, OOP Text-to-speech, Create a node for Text to image, Multimodal. [Backlog]: Age Estimator, Any CV Extractor, Face embedding.

#4. Characters: 459
Custom code and must be pinned on patch | - | Not started | 0% | pg (for postgrge execution) | Faldy Maulana Yuantoro |}\nFlow for CI/CD Flowise | - | - | - | - | - |}\nCredential MCP (for DB Access) | Might be on variable, credential, or any schema | - | - | - | - |}\n| Pilot Production release using CI/CD | - | - | - | - | - |}\n| TEMPI ATE COLLECTION | - | - | - | - | - | Collection of

Test Case C: DOCX Document

- **Filename:** Lampiran Teknis Arsitektur Flowise.docx
- **File Size:** 119 KB
- **Upload Timestamp:** 2025-12-29
- **Evidence of Indexing:**

←

Lampiran Teknis Arsitektur Flowise (LAMPIRAN TEKNIS ARSITEKTUR.docx)

LAMPIRAN TEKNIS ARSITEKTUR.docx

< Showing 1-1 of 1 chunks > 3,499 characters

#1. Characters: 3499
LAMPIRAN TEKNIS ARSITEKTUR PERANGKAT LUNAK

Nama Proyek: Agentic Development Platform & LLM Integration

Lingkungan: Staging Environment

Versi Dokumen: 1.0

Tanqaal: 23 Desember 2025

Test Case D: CSV Document

- **Filename:** OKR PAM Q4 2025.csv
- **File Size:** 12 KB
- **Upload Timestamp:** 2025-12-29
- **Evidence of Indexing:**

OKR PAM Q4 2025 (OKR 25Q4 - PAM PIA-PCS(FORM).csv)

Token Text Splitter

OKR 25Q4 - PAM PIA-PCS(FORM).csv

<

 Showing 1-24 of 24 chunks

>

🔍

 16,984 characters

#1. Characters: 250

Nama: NIK
_0:
_1:
_2:
GEDE DOKO HARIKUSUMA: 890030
_3:
Penting! Panduan dan informasi penggunaan fitur upload dapat dilihat pada sheet Example dan Legend.Kolom Objective tidak boleh di merge:
_4:

#2. Characters: 284

Nama: Posisi
_0:
_1:
_2:
GEDE DOKO HARIKUSUMA: MGR PLATFORM & INTEGRATION AUTOMATION
_3:
Penting! Panduan dan informasi penggunaan fitur upload dapat dilihat pada sheet Example dan Legend.Kolom Objective tidak boleh di merge:

#3. Characters: 248

Nama: Unit
_0:
_1:
_2:
GEDE DOKO HARIKUSUMA: PAM
_3:
Penting! Panduan dan informasi penggunaan fitur upload dapat dilihat pada sheet Example dan Legend.Kolom Objective tidak boleh di merge:
_4:

#4. Characters: 241

Nama:
_0:
_1:
_2:
GEDE DOKO HARIKUSUMA:
_3:
Penting! Panduan dan informasi penggunaan fitur upload dapat dilihat pada sheet Example dan Legend.Kolom Objective tidak boleh di merge:

#5. Characters: 399

Nama: No
_0: Kodifikasi
_1: Category OKR
_2: Objective
GEDE DOKO HARIKUSUMA: Key Result
_3: Definition of Done
Penting! Panduan dan informasi penggunaan fitur upload dapat dilihat pada sheet Example dan Legend.Kolom Objective tidak boleh di merge: Periode

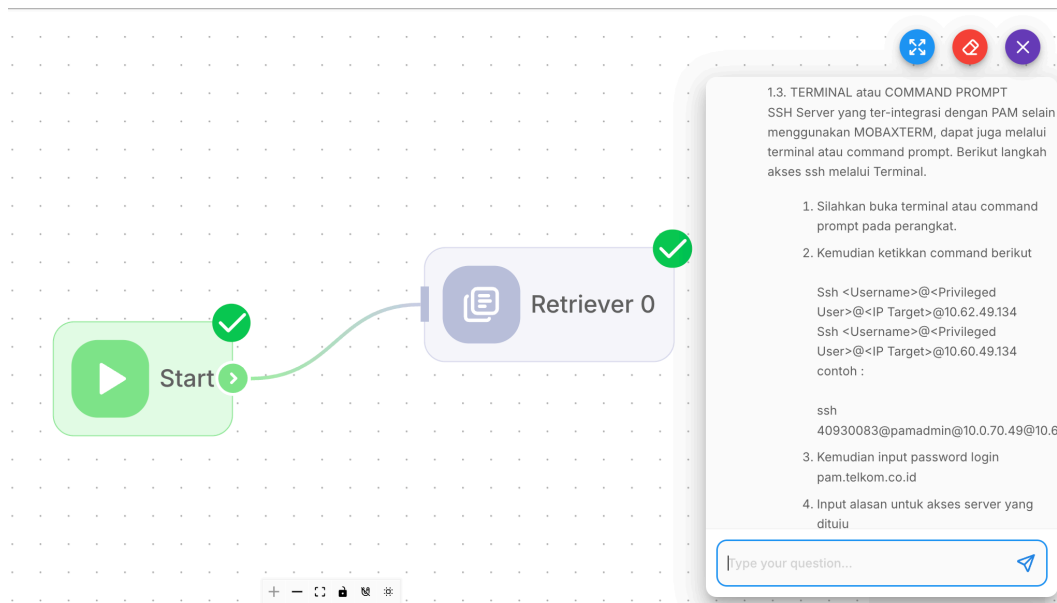
#6. Characters: 973

Nama: 1
_0: A.1
_1: Business Operational
_2: Ensure Operational Readiness & Support Fulfillment: 100% Fulfillment of Support Requests and Submission of Capacity Planning & RKAP Q1 2026
GEDE DOKO HARIKUSUMA: Membuktikan efektivitas AI coding assistant sebagai akselerator pengembangan aplikasi: 100%

4. Retrieval & RAG Verification (Evidence of Retrieval)

This section proves the AI can "read" the vector database.

- **User Question:** "Akses PAM dengan terminal atau command prompt"
- **Expected Source:** NewUserGuidePAMCyberArk-v3_3NNzU2NS.pdf
- **System Response:**



LAMPIRAN TEKNIS ARSITEKTUR PERANGKAT LUNAK

Nama Proyek: Agentic Development Platform & LLM Integration

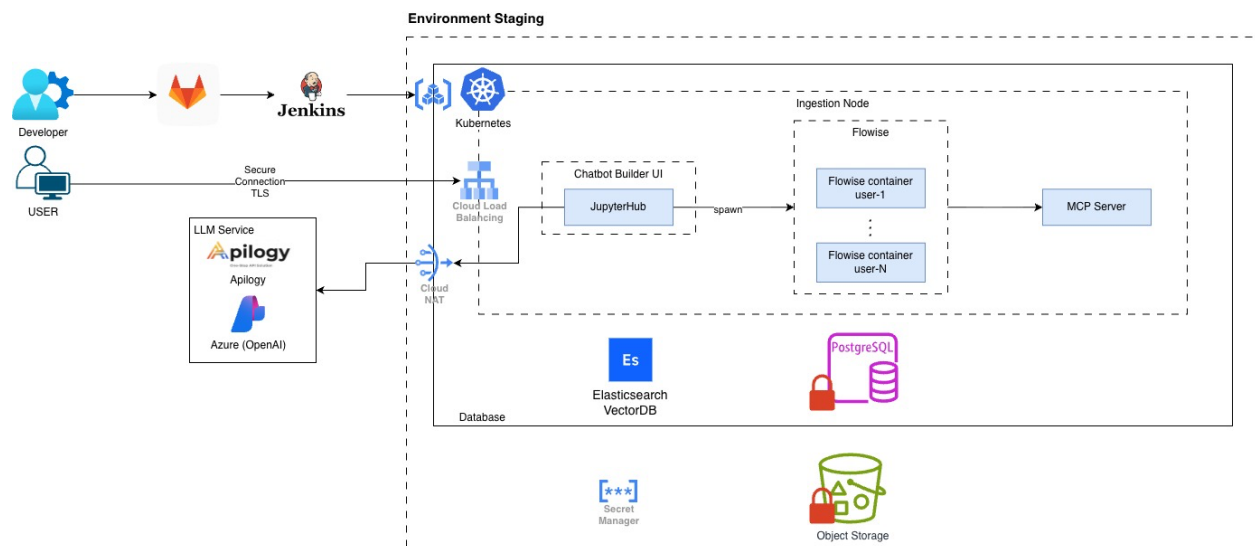
Lingkungan: Staging Environment

Versi Dokumen: 1.0

Tanggal: 23 Desember 2025

1. ARSITEKTUR SISTEM (SAD)

Bagian ini menggambarkan topologi arsitektur sistem pada lingkungan *Staging*. Sistem dirancang menggunakan prinsip *microservices* yang berjalan di atas platform Kubernetes.



2. DESKRIPSI KOMPONEN DAN LAYANAN

Sistem ini terdiri dari beberapa *layer* infrastruktur dan aplikasi. Berikut adalah rincian teknis setiap komponen:

2.1 DevOps & CI/CD Pipeline

Komponen untuk manajemen siklus pengembangan perangkat lunak.

Komponen	Fungsi Utama	Keterangan
GitLab	Source Code Management (SCM)	Repository utama untuk penyimpanan kode dan <i>version control</i> .
Jenkins	CI/CD Automation	Server otomatisasi yang melakukan <i>build image</i> , pengujian, dan <i>deployment</i> ke Kubernetes.

2.2 Infrastruktur Cloud & Network

Layanan dasar yang menangani orkestrasi, konektivitas, dan keamanan jaringan.

Komponen	Fungsi Utama	Keterangan
Kubernetes (K3s)	Container Orchestration	Menjalankan seluruh <i>workload</i> aplikasi (Flowise, JupyterHub, dll).
Cloud Load Balancing	Ingress Gateway	Pintu gerbang trafik masuk, menangani terminasi SSL/TLS.
Cloud NAT	Outbound Connectivity	Memfasilitasi akses internet aman untuk layanan internal (private IP) menuju API eksternal.
Secret Manager (vault)	Security	Penyimpanan terenkripsi untuk API Keys, kredensial database, dan sertifikat.

2.3 Komponen Aplikasi (Workload)

Inti logika bisnis dari platform *Chatbot Builder*.

Komponen	Fungsi Utama	Keterangan
JupyterHub	User Interface Spawner	Bertindak sebagai UI awal dan <i>spawner</i> yang memicu pembuatan container per pengguna.
Flowise	Chatbot App Builder	<i>Low-code tool</i> untuk membangun alur chatbot. Berjalan dalam mode <i>multi-tenant</i> (1 container per user).
MCP Server	RDBMS connector	<i>Model Context Protocol</i> server untuk menjembatani Flowise dengan RDBMS PostgreSQL dan MySQL eksternal.

2.4 Manajemen Data (Persistence)

Penyimpanan data terstruktur dan vektor.

Komponen	Fungsi Utama	Keterangan
Elasticsearch	Vector Database	Menyimpan <i>embedding</i> data untuk fitur <i>Retrieval-Augmented Generation</i> (RAG).
PostgreSQL	Relational Database	Menyimpan data user, sesi, konfigurasi flow, dan log transaksi.
Object Storage	File Storage	Penyimpanan untuk aset statis dan dokumen yang diunggah pengguna (S3 Compatible).

2.5 Layanan Eksternal (LLM Service)

Integrasi LLM Service Provider.

Komponen	Fungsi Utama	Keterangan
Apilogy	Aggregator API	Layanan solusi API terintegrasi.
Azure OpenAI	LLM Provider	Penyedia model LLM (GPT-3.5/GPT-4) kelas <i>enterprise</i> .

3. ALUR KERJA SISTEM

Mekanisme operasional sistem dari sisi *deployment* hingga interaksi pengguna:

1. **Deployment (CI/CD):** Kode yang diperbarui di *GitLab* akan diproses oleh *Jenkins* untuk kemudian diperbarui di *cluster Kubernetes*.
2. **Akses Pengguna:** Pengguna mengakses sistem melalui protokol HTTPS/TLS. Trafik didistribusikan oleh *Load Balancer* menuju *JupyterHub*.
3. **Inisialisasi Sesi:** *JupyterHub* membuat (*spawn*) kontainer *Flowise* yang terisolasi khusus untuk pengguna tersebut.
4. **Chatbot:**
 - Interaksi chat dikelola oleh *Flowise*.
 - Pencarian konteks data dilakukan ke *Elasticsearch*.
 - Inferensi AI dilakukan dengan memanggil *Azure OpenAI* atau *Apilogy*.
 - Koneksi dengan RDBMS eksternal (PostgreSQL dan MySQL) oleh *Flowise* melalui MCP Server.
5. **Storage:** Seluruh konfigurasi dan riwayat disimpan secara persisten di *PostgreSQL*.